Problem Set 2 — Answer Key

Origins of Mathematics, Fall 2026

Grading note. A proof must justify the general case; numerical examples alone earn no credit for "prove" parts. Diagrams must be labeled.

1 Squaring numbers ending in 5

(a) Conjecture

If n is a positive integer whose decimal representation ends in 5, write $n = 10k + 5$ with $k \geq 0$. Then

$$n^2 = 100\,k(k+1) + 25.$$

In words: *delete the final 5, multiply the remaining number by the next integer, and append 25.*

(b) Four examples

$$55^2 = 3025, \quad 65^2 = 4225, \quad 75^2 = 5625, \quad 85^2 = 7225.$$

(Check: $5 \cdot 6 = 30$, $6 \cdot 7 = 42$, $7 \cdot 8 = 56$, $8 \cdot 9 = 72$, each followed by 25.)

(c) Proof

Let $n = 10k + 5$. Then

$$n^2 = (10k+5)^2 = 100k^2 + 100k + 25 = 100\,k(k+1) + 25.$$

Since $k(k+1)$ is an integer, $100\,k(k+1)$ occupies the decimal places from the hundreds upward, and the added 25 occupies exactly the tens and units places. Hence the decimal representation of n^2 is the decimal representation of $k(k+1)$ followed by the two digits "25." (When $k = 0$, $n = 5$ and $n^2 = 25$, consistent with the empty prefix.) □

2 Finding a pair

Partition $\{1, 2, \ldots, 8\}$ into the four pairs

$$\{1, 8\}, \quad \{2, 7\}, \quad \{3, 6\}, \quad \{4, 5\},$$

each of which sums to 9.

(a) Proof

Choosing five distinct numbers from a set partitioned into four pairs means, by the **Pigeonhole Principle**, that two of the chosen numbers lie in the same pair. Any two members of a pair $\{j, 9 - j\}$ sum to 9. Hence some two chosen numbers add to 9. □

(b) Four numbers with no pair summing to 9

Yes. Example: $\{1, 2, 3, 4\}$. No two of these sum to 9 (the largest possible sum is $3 + 4 = 7$). More generally, choosing at most one element from each of the four pairs yields a valid set of size 4; e.g. $\{1, 2, 3, 4\}$, $\{1, 2, 3, 5\}$, $\{1, 2, 6, 7\}$, etc.

Sharpness. Four is the maximum: by part (a), any five numbers force a pair summing to 9.

3 A remainder problem

We seek the smallest $n \in \mathbb{N}$ with

$$n \equiv 1 \pmod 2, \qquad n \equiv 4 \pmod 5, \qquad n \equiv 8 \pmod{11}.$$

The moduli $2, 5, 11$ are pairwise coprime, so the Chinese Remainder Theorem guarantees a unique solution modulo $M = 2 \cdot 5 \cdot 11 = 110$.

(a) Sun Zi reconstruction

Step 1 (unit contributions). For each modulus m_i, set $M_i = M/m_i$ and find y_i with $M_i y_i \equiv 1 \pmod{m_i}$:

$$\begin{aligned}
m_1 = 2: &\quad M_1 = 55, \quad 55y_1 \equiv 1 \pmod 2 \iff y_1 \equiv 1 \pmod 2, \quad y_1 = 1.\\
m_2 = 5: &\quad M_2 = 22, \quad 22y_2 \equiv 1 \pmod 5 \iff 2y_2 \equiv 1 \pmod 5, \quad y_2 = 3.\\
m_3 = 11: &\quad M_3 = 10, \quad 10y_3 \equiv 1 \pmod{11} \iff -y_3 \equiv 1 \pmod{11}, \quad y_3 = 10.
\end{aligned}$$

(Check: $55 \cdot 1 = 55 \equiv 1 \pmod 2$; $22 \cdot 3 = 66 \equiv 1 \pmod 5$; $10 \cdot 10 = 100 \equiv 1 \pmod{11}$.)

Step 2 (combine with remainders). The reconstruction formula gives

$$n \equiv \sum_i r_i M_i y_i \pmod M = 1 \cdot (55 \cdot 1) + 4 \cdot (22 \cdot 3) + 8 \cdot (10 \cdot 10) \pmod{110}.$$

Compute:

$$55 + 4 \cdot 66 + 8 \cdot 100 = 55 + 264 + 800 = 1119.$$

Reduce modulo 110:

$$1119 = 10 \cdot 110 + 19 \implies n \equiv 19 \pmod{110}.$$

The smallest positive solution is $\boxed{n = 19}$.

(b) Verification

$$19 = 2 \cdot 9 + 1 \checkmark, \qquad 19 = 5 \cdot 3 + 4 \checkmark, \qquad 19 = 11 \cdot 1 + 8 \checkmark.$$

(c) All solutions

Since the moduli are pairwise coprime, CRT gives a unique residue class modulo 110. Hence the full solution set is

$$\boxed{n = 19 + 110t, \quad t = 0, 1, 2, \ldots}$$

Why this is complete: if n and n' are two solutions, then $n - n'$ is divisible by each of $2, 5, 11$, hence by their product 110 (coprimality). So any two solutions differ by a multiple of 110, and every number of the form $19 + 110t$ satisfies all three congruences. $\square$

4 Trisecting a line segment

(a) Construction

Given segment AB.

Step 1. Draw a ray ℓ from A at any convenient angle to AB (not collinear with AB).

Step 2. With compass set to any convenient radius r, strike off three equal segments along ℓ from A: obtain points P_1, P_2, P_3 with

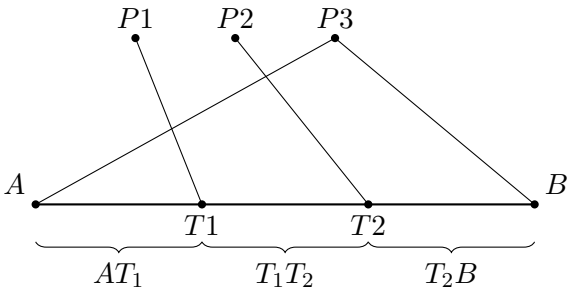
$$AP_1 = P_1P_2 = P_2P_3 = r.$$

Step 3. Draw line P_3B.

Step 4. Through P_1 and P_2, construct lines parallel to P_3B (copy the angle $\angle AP_3B$ at P_1 and P_2; Euclid I.23/I.31).

Step 5. Let the parallel through P_1 meet AB at T_1, and the parallel through P_2 meet AB at T_2.

Result: T_1, T_2 trisect AB.



(b) Proof

Apply **Euclid VI.2** to triangle AP_3B. Since $P_1T_1 \parallel P_3B$,

$$\frac{AT_1}{T_1P_3} = \frac{AP_1}{P_1P_3} = \frac{r}{2r} = \frac{1}{2}.$$

Equivalently, $AT_1 : T_1B = 1 : 2$ in triangle ABP_3 with $P_1T_1 \parallel P_3B$, so $AT_1 = \frac{1}{3}AB$.

Similarly, $P_2T_2 \parallel P_3B$ gives $AT_2 : T_2B = AP_2 : P_2P_3 = 2r : r = 2 : 1$, so $AT_2 = \frac{2}{3}AB$.

Therefore

$$T_1T_2 = AT_2 - AT_1 = \tfrac{2}{3}AB - \tfrac{1}{3}AB = \tfrac{1}{3}AB,$$

and $T_2B = AB - AT_2 = \frac{1}{3}AB$. Hence

$$AT_1 = T_1T_2 = T_2B = \tfrac{1}{3}AB. \quad \square$$

Optional extension

To divide AB into n equal parts: lay off n equal segments $AP_1 = P_1P_2 = \cdots = P_{n-1}P_n$ along the auxiliary ray, draw P_nB, and through each P_i $(1 \le i \le n-1)$ draw the parallel to P_nB. By VI.2, the intersections on AB cut it in the ratio $1 : 1 : \cdots : 1$, i.e. into n equal parts.

5 One quadratic, three methods

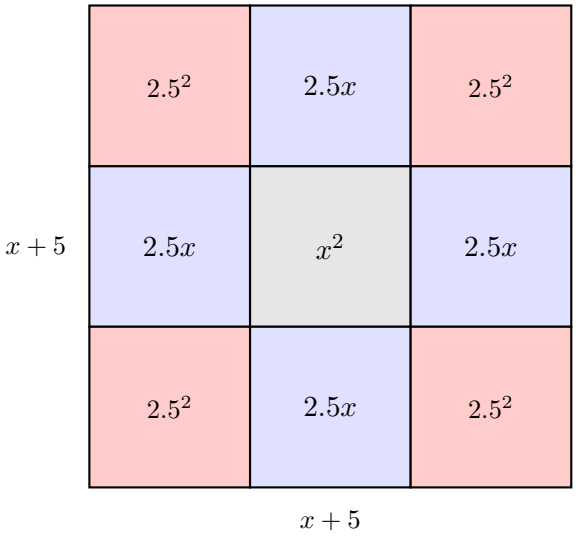
Equation: $x^2 + 10x = 39$.

(a) Geometry (four-rectangle completion)

Begin with a square of side x (area x^2). The term $10x$ is represented by four congruent rectangles, one attached to each side of the square. Since $10x = 4 \cdot (2.5x)$, each rectangle has dimensions

$$x \times \tfrac{10}{4} = x \times 2.5,$$

so each has area $2.5x$, and the four together have area $10x$.



The four *missing corner squares* each have side 2.5, hence area $2.5^2 = 6.25$. Total area to add:

$$4 \cdot 6.25 = 25.$$

Adding this to both sides of $x^2 + 10x = 39$:

$$x^2 + 10x + 25 = 39 + 25 = 64.$$

The left side is the area of the completed square of side $x + 5$:

$$(x + 5)^2 = 64 \implies x + 5 = 8 \implies x = 3.$$

(Lengths are positive, so we take the positive square root.)

(b) Quadratic formula

$$x^2 + 10x - 39 = 0 \implies x = \frac{-10 \pm \sqrt{100 + 156}}{2} = \frac{-10 \pm \sqrt{256}}{2} = \frac{-10 \pm 16}{2}.$$

Thus

$$x_1 = \frac{6}{2} = 3, \qquad x_2 = \frac{-26}{2} = -13.$$

(c) Average and product

Sum of roots $= -10$, product $= -39$. Average:

$$\bar{x} = \frac{3 + (-13)}{2} = -5.$$

Write the roots as $\bar{x} \pm d = -5 \pm d$. Their product is

$$(-5 + d)(-5 - d) = 25 - d^2 = -39 \implies d^2 = 64 \implies d = 8.$$

Hence the roots are $-5 + 8 = 3$ and $-5 - 8 = -13$.

(d) Comparison

All three methods agree: roots 3 and -13.

- **Geometry** yields only $x = 3$ because it interprets x as a *length*; the completed square has side $x + 5 = 8 > 0$, and a negative side length is meaningless. The construction automatically discards $x = -13$ (which would give side $x + 5 = -8$, impossible for a geometric square).
- **Verification:**

$$3^2 + 10 \cdot 3 = 9 + 30 = 39 \checkmark, \qquad (-13)^2 + 10(-13) = 169 - 130 = 39 \checkmark.$$

The geometric method is a *restricted* solution method (positive roots only); the algebraic methods recover the full solution set over $\mathbb{R}$.

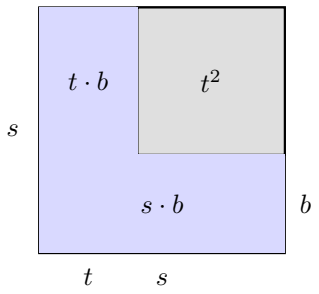
Optional challenge: A geometric identity

Let $a > b > 0$. We prove

$$ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2.$$

Setup. Let $s = \frac{a+b}{2}$ (the average) and $t = \frac{a-b}{2}$ (the half-difference). Consider the square of side s. Inside it, place the square of side t in the top-right corner. The *gnomon* is the L-shaped region that remains:

$$\text{Area of gnomon} = s^2 - t^2 = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2.$$



Why the gnomon has area ab. The gnomon can be split into two non-overlapping rectangles:

1. A bottom rectangle of width s and height $s - t = b$. Its area is $s \cdot b$.

2. A left rectangle of width t and height $s - t = b$. Its area is $t \cdot b$.

Together, their total area is

$$s \cdot b + t \cdot b = (s + t) \cdot b.$$

But recall our definitions: $s = \frac{a+b}{2}$ and $t = \frac{a-b}{2}$. Their sum is exactly a:

$$s + t = \frac{a + b}{2} + \frac{a - b}{2} = \frac{2a}{2} = a.$$

Therefore, the total area of the gnomon is $a \cdot b = ab$.
Conclusion. The same gnomon has area $s^2 - t^2$ (as outer square minus inner square) and area ab (as the two rectangles). Hence

$$ab = \left(\frac{a + b}{2}\right)^2 - \left(\frac{a - b}{2}\right)^2. \quad \square$$

Interpretation. The identity says: *product = (average)^2 − (half-difference)^2*. Equivalently, the product is the square of the average diminished by the square of the half-difference. This is exactly the mechanism behind part 5(c): given the average $\bar{x}$ and product P, the half-difference d satisfies

$$P = \bar{x}^2 - d^2 \implies d = \sqrt{\bar{x}^2 - P},$$

and the roots are $\bar{x} \pm d$. The gnomon identity is the geometric form of "sum-and-product determines the roots."